

INPUT

Travel demand

Regional & inter-regional
origin-destination trip data

Vehicle types | drivetrain technologies | fuels

Graph topology

Nodes (regions) | edges (transport
connections) | route definitions

Parameters on charging infrastructure expansion

Differentiated by fuel & charging power
| Maximum possible speed of build-up

Fuel prices | technology costs |
charging infrastructure costs | route-
related / location-specific costs...

Optimization model

- Linear program
- System cost-minimization
- Network-based transport demand coverage
- Vehicle stock modeling
- Sizing of charging infrastructure
- Spatially flexible charging activity

OUTPUT

Trips per vehicle type
| technology | fuel

Vehicle stock shift

Charging demand
by location

Charging infrastructure
expansion